

HOMEWORK 4
BIOSTATISTICS 794
DUE OCTOBER 23TH, 2026

Instructions

- Submit a single .Qmd or .html/.pdf file with code, figures, and interpretations.
- Clearly label each section and question.
- Use cross-validation wherever possible, with appropriate metrics.
- Comment your R code and explain your interpretations in plain language.

1. **[14 points]** For this question, use the **Heart Disease** dataset (`mlbench::Heart`) or an equivalent from the UCI repository.
 - (a) (2 pts) Explore the dataset and describe the variables (which are continuous, categorical, or binary?).
 - (b) (1 pt) Split the data into training (80%) and testing (20%) sets.
 - (c) (3 pts) Fit a **Naive Bayes** classifier using all predictors. Use `train(..., method="naive.bayes")` from the `caret` package.
 - (d) (3 pts) Evaluate AUC, accuracy, sensitivity, specificity, PPV, and NPV on the test set.
 - (e) (3 pts) Repeat the analysis after converting some variables to categorical (e.g., bin cholesterol or age). Compare performance.
 - (f) (2 pts) Briefly discuss when Naive Bayes is advantageous in public health prediction.
2. **[16 points]** Use the **birthwt** dataset from the `MASS` package.
 - (a) (2 pts) Describe the data and the outcome variable (`bwt`).
 - (b) (3 pts) Fit a **CART regression tree** using `rpart(bwt ~ ., data=birthwt)`.
 - (c) (3 pts) Use cross-validation to select the best complexity parameter (`cp`).
 - (d) (3 pts) Plot the tree and interpret the top two splits in the context of maternal health.
 - (e) (3 pts) Compare model RMSE to a standard multiple linear regression model.
 - (f) (2 pts) Discuss how overfitting might appear in a regression tree.
3. **[22 points]** Use the **NHANES** dataset (via `NHANES` package) that we used in the dimension reduction example. Choose a continuous outcome such as BMI or one of the health biomarkers (e.g., blood pressure, cholesterol, etc.).

When cleaning the data remove duplicate IDs, only keep variables with at most 20% missingness, and then drop all observations with missing values. To do this, you can use the following code:

```

nhanes <- NHANES |>
clean_names() |>
distinct(id, .keep_all = TRUE) |>
select(where(~ mean(is.na(.x)) <= 0.20)) |>
drop_na()

```

Note: at this point you will have to do some more data cleaning and may want to remove some redundant rows.

- (a) (3 pts) Split the data into training (70%) and test (30%) datasets. Preprocess the data: handle missing values, and normalize predictors using `recipes`.
 - (b) (2 pts) Fit a **Random Forest** regression model using `randomForest` (or `ranger`) for a fixed `ntree` and `mtry`.
 - (c) (3 pts) Tune `mtry` using cross-validation.
 - (d) (3 pts) Plot variable importance. Identify the top three predictors and interpret them.
 - (e) (4 pts) Create and interpret a **partial dependence plot (PDP)** for one important variable.
 - (f) (3 pts) Use a lasso, ridge, or other penalization method to the data.
 - (g) (4 pts) Compare the test RMSE for the random forest and the penalization regression model.
4. [36 points] Use the **Alzheimer's disease** data (`ad_data` in the `modeldata` package). The outcome `Class` indicates whether a patient is cognitively Impaired or a Control. The predictors are 130 measurements, mostly plasma protein levels, along with age, male, Genotype (APOE), and cerebrospinal fluid markers (`tau`, `p_tau`, `Ab_42`). Fit all neural networks in this question using `torch` and `luz`, following the first example in the Neural Networks II notes (Framingham).
- (a) (5 pts) Describe the data (sample size, class balance, predictor types). Split the data into training (60%), validation (20%), and test (20%) sets, stratified on `Class`. Using `recipes`, create dummy variables for categorical predictors and normalize the numeric predictors, learning the preprocessing on the training set only. Convert the data to `torch` tensors and `dataloaders`.
 - (b) (5 pts) Write a function (e.g., using `nn_module`) that builds a feed-forward network with a flexible number of hidden layers and units and optional dropout, and a function that fits it with `luz` using the Adam optimizer, weight decay, and early stopping on the validation loss.
 - (c) (7 pts) Tune the network over a grid that includes at least: (i) 4 or more architectures (e.g., one hidden layer with 8 or 32 units, and two- and three-layer networks), (ii) dropout probability (e.g., 0 and 0.3), (iii) weight decay (e.g., 0 and 10^{-3}), and (iv) learning rate (e.g., 0.01 and 0.001). Record the training AUC,

validation AUC, and number of epochs run for every configuration. Because the grid may be slow to run, save the results to disk and reload them when re-rendering.

- (d) (6 pts) Present the top 10 configurations ranked by validation AUC, and plot validation AUC across architectures and regularization settings. Compare training and validation AUC: which configurations overfit, and what effect do dropout and weight decay have? Note that the validation set is small, so comment on which configurations are effectively tied and which model you would choose.
 - (e) (6 pts) Fit a logistic regression with a lasso penalty (`glmnet`) and a random forest (e.g., `caret` with tuned `mtry`) using the same training data, tuning each on the training data only (or on the validation set). Choose a classification threshold for each of the three models (network, lasso, random forest) on the validation set, for example by maximizing Youden's J .
 - (f) (4 pts) Evaluate all three final models once on the test set, reporting AUC, accuracy, sensitivity, and specificity at the chosen thresholds. Which model do you recommend?
 - (g) (3 pts) Discuss: With only a few hundred patients and 130 predictors, what are the risks of using a neural network here? When might neural networks be unnecessary or overkill for public health data?
5. [12 points] You are studying a dataset of 500 patients, each with 200 biomarkers measured at baseline. You plan to visualize the relationships between individuals using a low-dimensional embedding (2D or 3D). Consider the following situations and state whether a **global** or **local** dimension reduction method would be more appropriate. Justify your answer in 2–3 sentences.
- (a) (3 pts) You believe there is a single underlying continuous “disease severity” score that explains most of the biomarker variation.
 - (b) (3 pts) You are interested in identifying small subgroups of patients with very similar biomarker profiles but distinct from the rest of the population.
 - (c) (3 pts) You plan to create a 2D visualization of single-cell RNA expression data to identify cell-type clusters.
 - (d) (3 pts) You have time-series physiological data (e.g., heart rate, sleep, and activity) and want to summarize the overall level and variability across participants.